Lec.1 Bacterial Growth

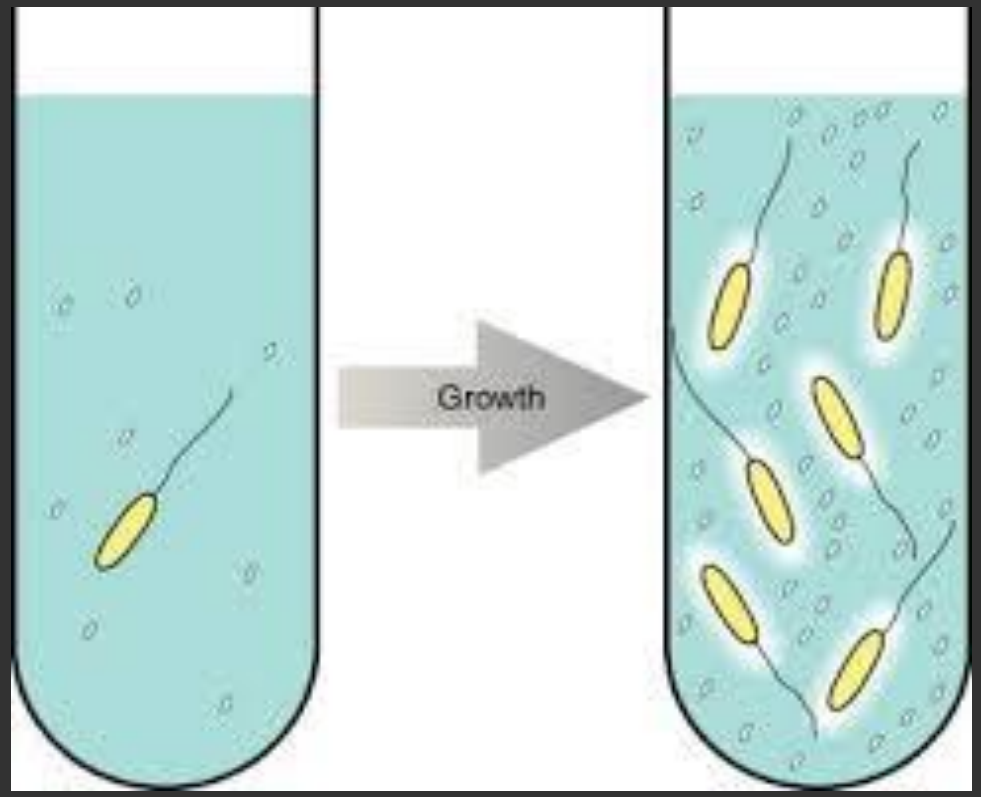
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BACTERIAL GROWTH



Growth of Microbes

- Increase in *number* of cells, not cell size
- One cell becomes **colony** of millions of cells

- **Bacterial Colony is a cluster of** millions of cells



Control of growth is important for

- Infection control
- Growth of industrial and biotech organisms



Bacterial growth

- Binary fission
- Generation time
- Phases of growth

Binary fission

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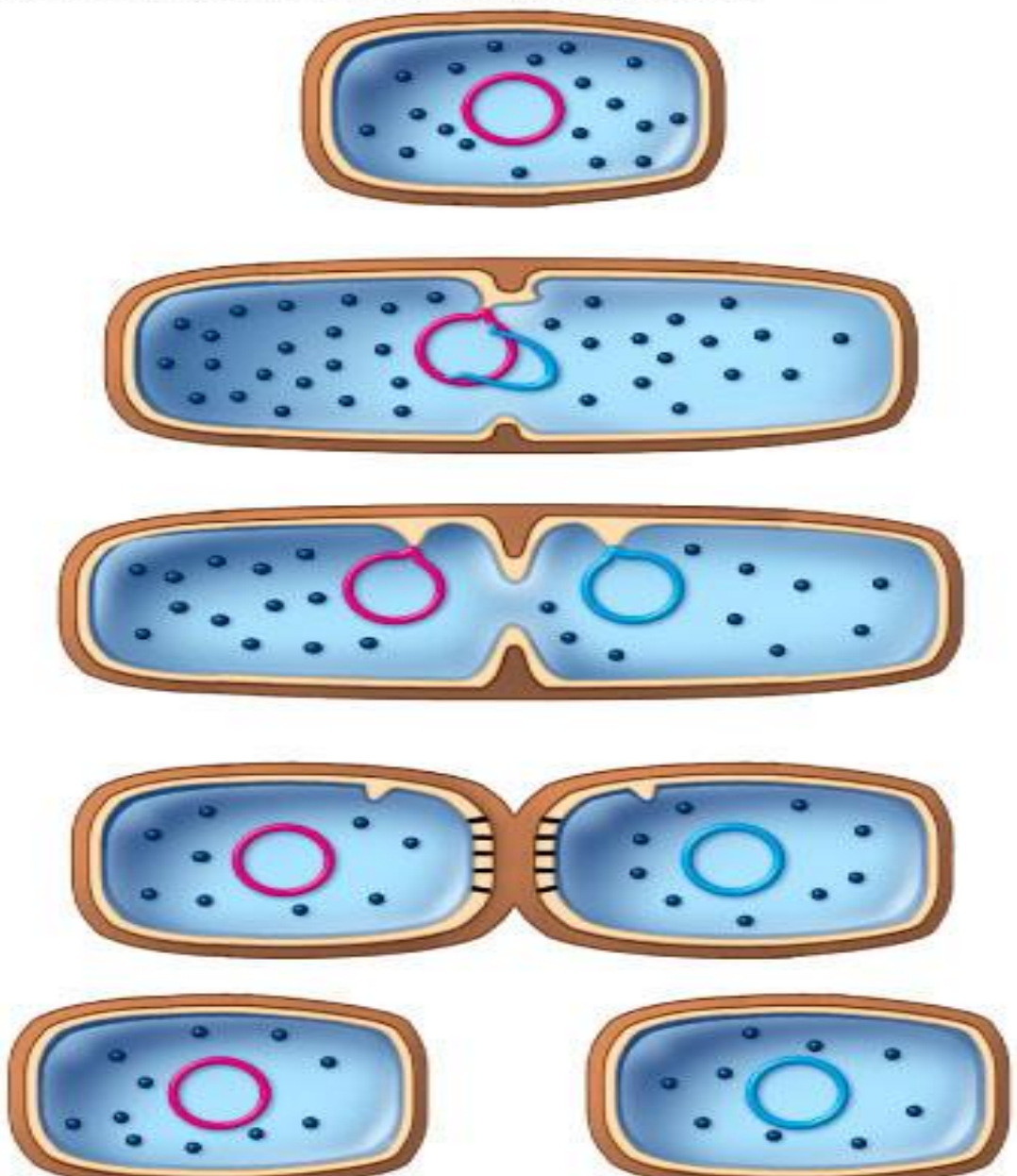
(a) A young cell at early phase of cycle.

(b) A parent cell prepares for division by enlarging its cell wall, cell membrane, and overall volume. Midway in the cell, the wall develops notches that will eventually form the transverse septum, and the duplicated chromosome becomes affixed to a special membrane site.

(c) The septum wall grows inward, and the chromosomes are pulled toward opposite cell ends as the membrane enlarges. Other cytoplasmic components are distributed (randomly) to the two developing cells.

(d) The septum is synthesized completely through the cell center, and the cell membrane patches itself so that there are two separate cell chambers.

(e) At this point, the daughter cells are divided. Some species will separate completely as shown here, while others will remain attached, forming chains or doublets, for example



Generation time

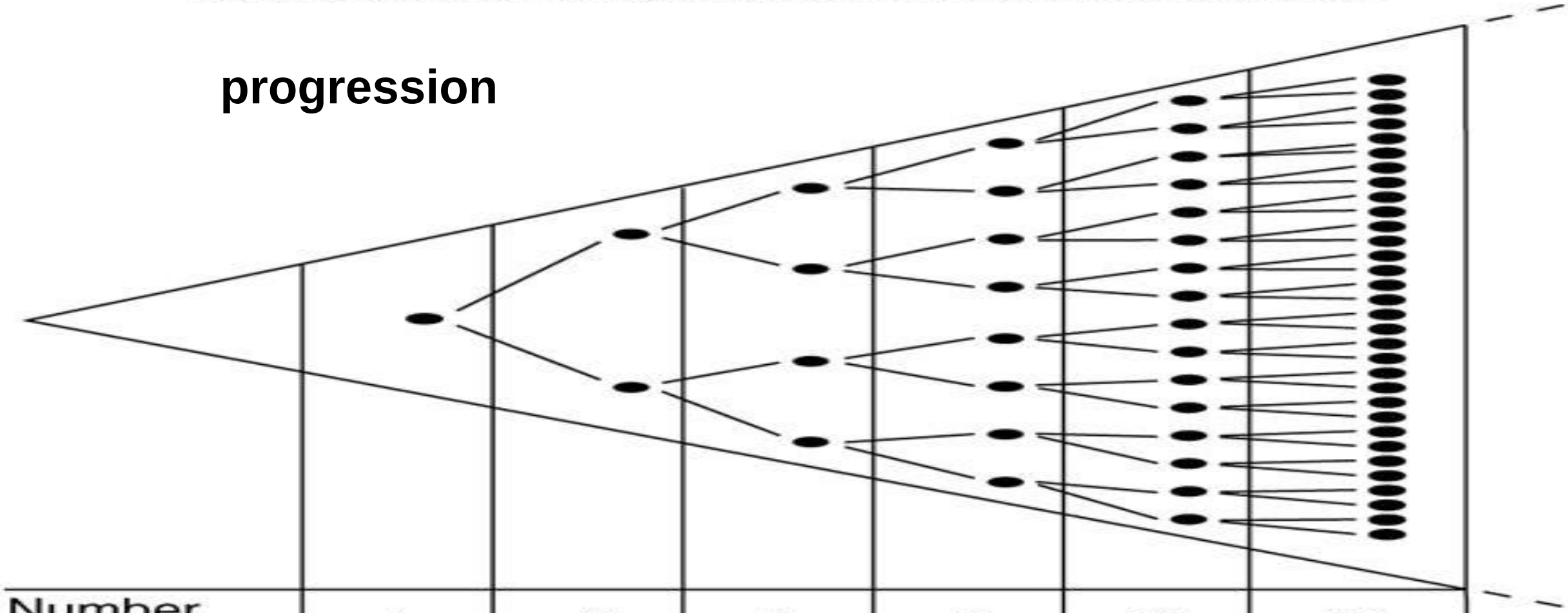
- The time required to for a population to double (doubling time) in number.
- Ex. *Escherichia coli* (*E. coli*) double every 20 minutes
- Ex. *Mycobacterium tuberculosis* double every 12 to 24 hours

Principles of Bacterial Growth

- Growth can be calculated
  - $N_t = N_0 \times 2^n$
    - *(N_t) number of cells in population*
    - *(N_0) original number of cells in the population*
    - *(n) number of divisions*
    - Example
      - N_0 = 10 cells in original population
      - n = 12
        - 4 hours assuming 20 minute generation time
      - $N_t = 10 \times 2^{12}$
      - $N_t = 10 \times 4{,}096$
      - $N_t = 40{,}960$

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progression



Number of cells	1	2	4	8	16	32
Number of generations		1	2	3	4	5
Exponential value		2^1 (2×1)	2^2 (2×2)	2^3 (2×2×2)	2^4 (2×2×2×2)	2^5 (2×2×2×2×2)

Growth in Batch Culture

Bacteria reproduce by binary fission, a process by which one parent cell divides to form two progeny cells.

Bacteria growing in batch culture produce a **growth curve** with up to four distinct phases. Batch cultures are grown in tubes or flasks and are **closed systems** where no fresh nutrients are added or waste products removed.

1. **Lag phase** which vigorous metabolic activity or adaptation occurs but cells do not divide. For example, with a nutritionally poor medium, several anabolic pathways need to be turned on, resulting in a lag before active growth begins.

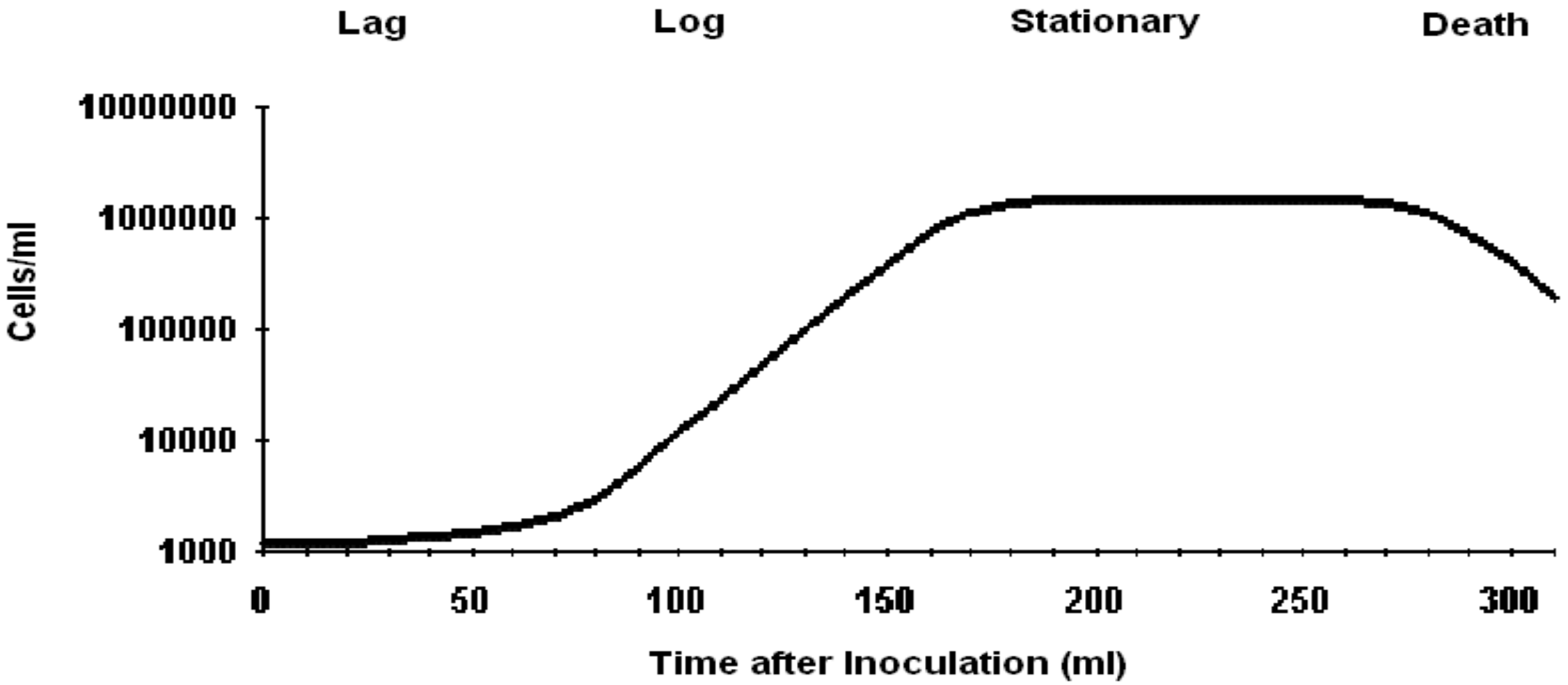
2. In **log** or **exponential phase**, the cells are growing as fast as they can, limited only by growth conditions and genetic potential. During this phase, almost all cells are alive, they are most nearly identical, and they are most affected by outside influences like disinfectants.

3. **Stationary phase** Due to nutrient depletion and/or accumulation of toxic end products (antibiotics, toxins bacteriocins, and acids…….). Replication stops and cells enter a **stationary phase** where there is no net change in cell number. Growth rate of cells =dead rate of cells

4. **Death phase** occurs when cells can no longer maintain viability and numbers decrease as

a proportion.

Growth in Batch Culture



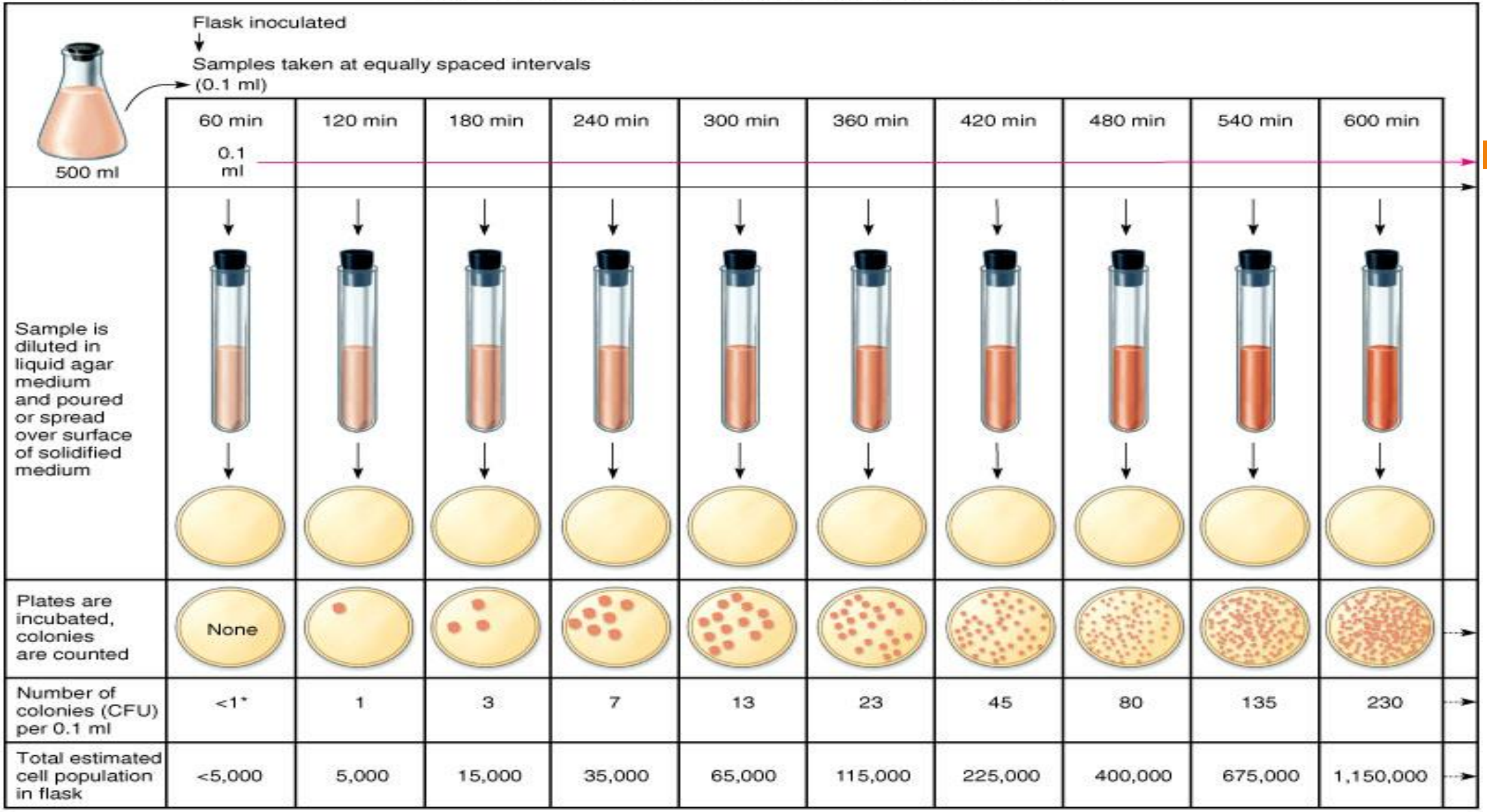
Mean Generation Time and Growth Rate

- The mean generation time (doubling time) is the amount of time required for the concentration of cells to double during the log stage. It is expressed in units of minutes.
- Growth rate (min^{-1}) = $\frac{1}{\text{mean generation time}}$
- Mean generation time can be determined directly from a semilog plot of bacterial concentration vs time after inoculation

Measuring Growth

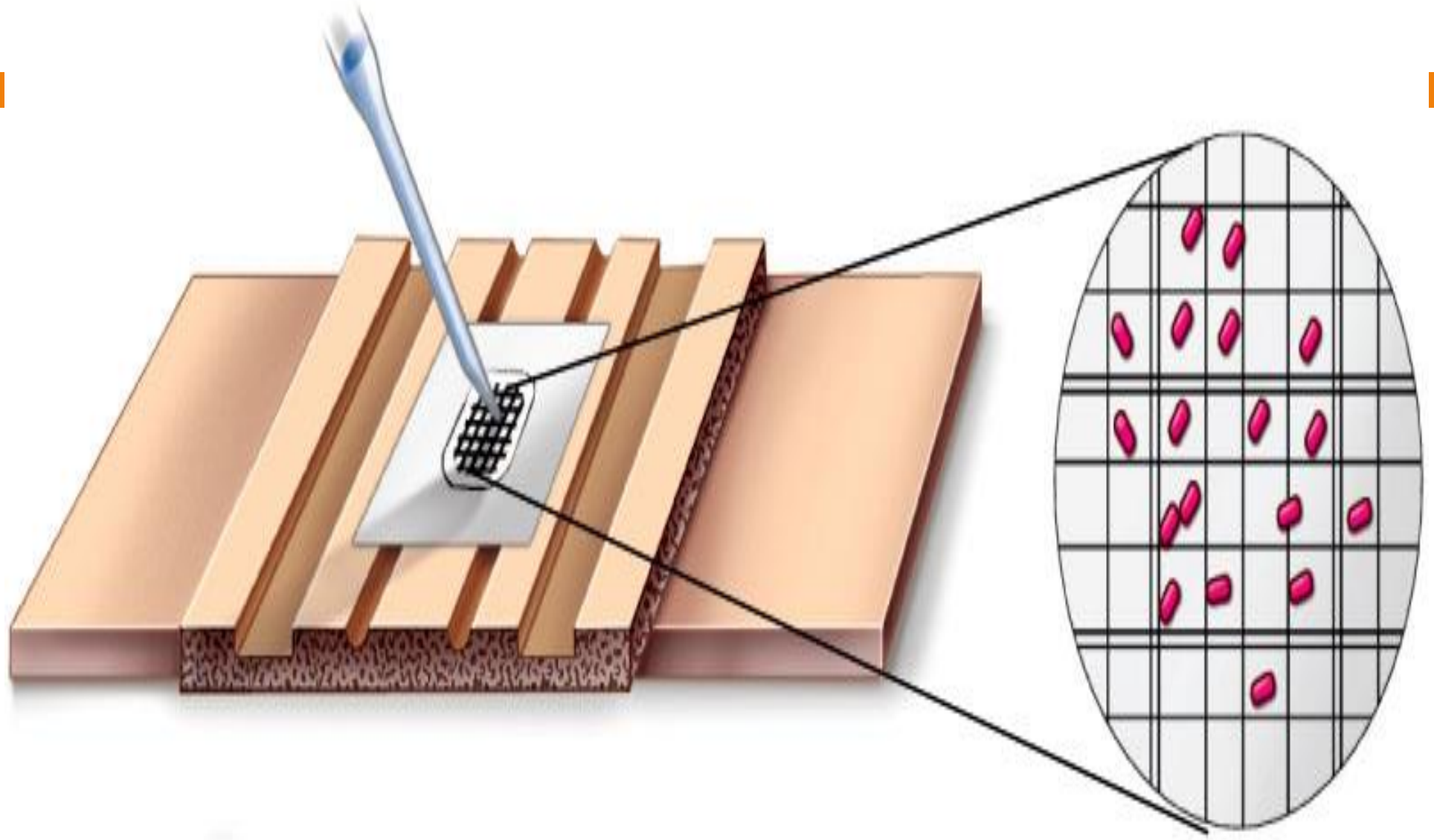
- Direct methods – count individual cells
- Indirect Methods – measure effects of bacterial growth

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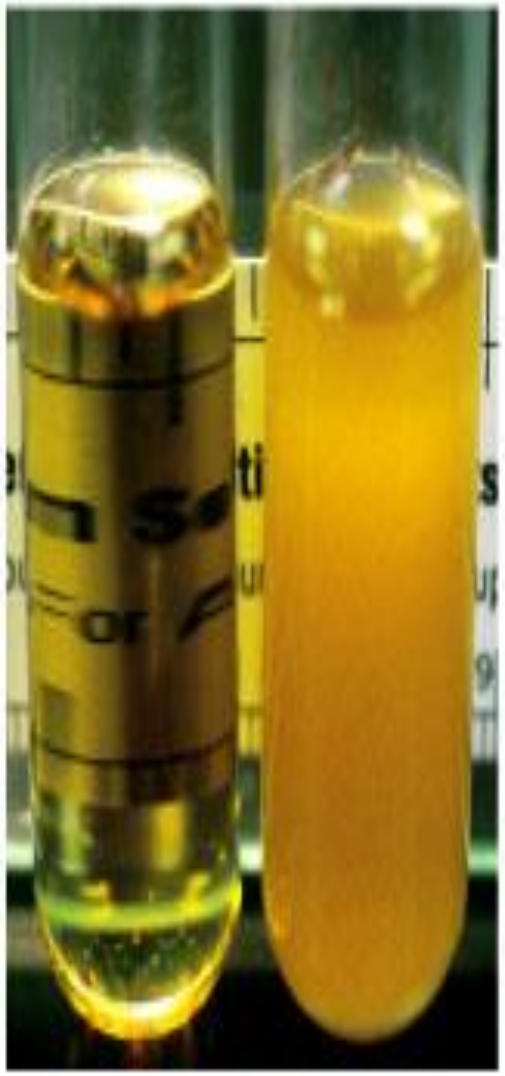


* Only means that too few cells are present to be assayed.

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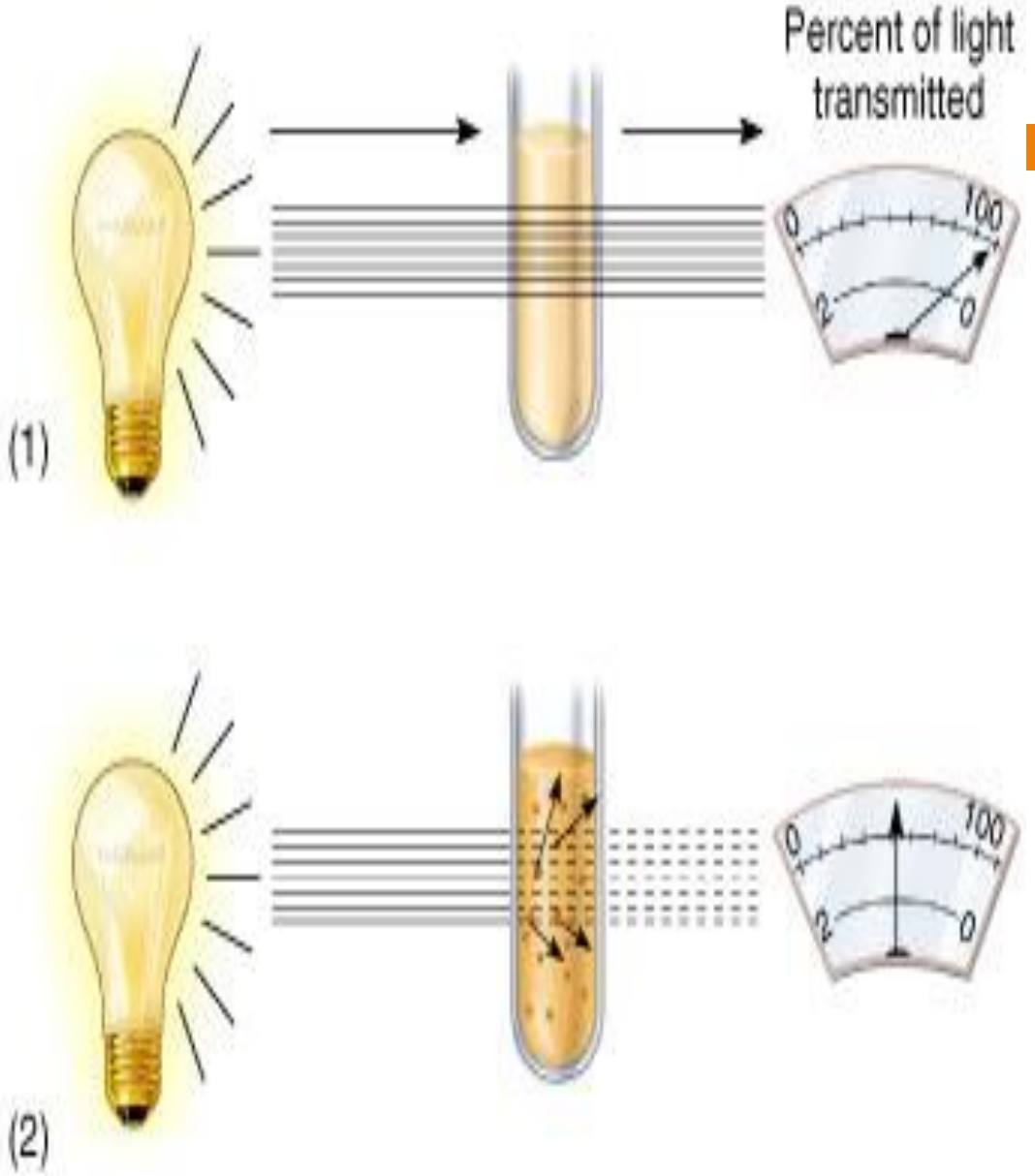
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(a)

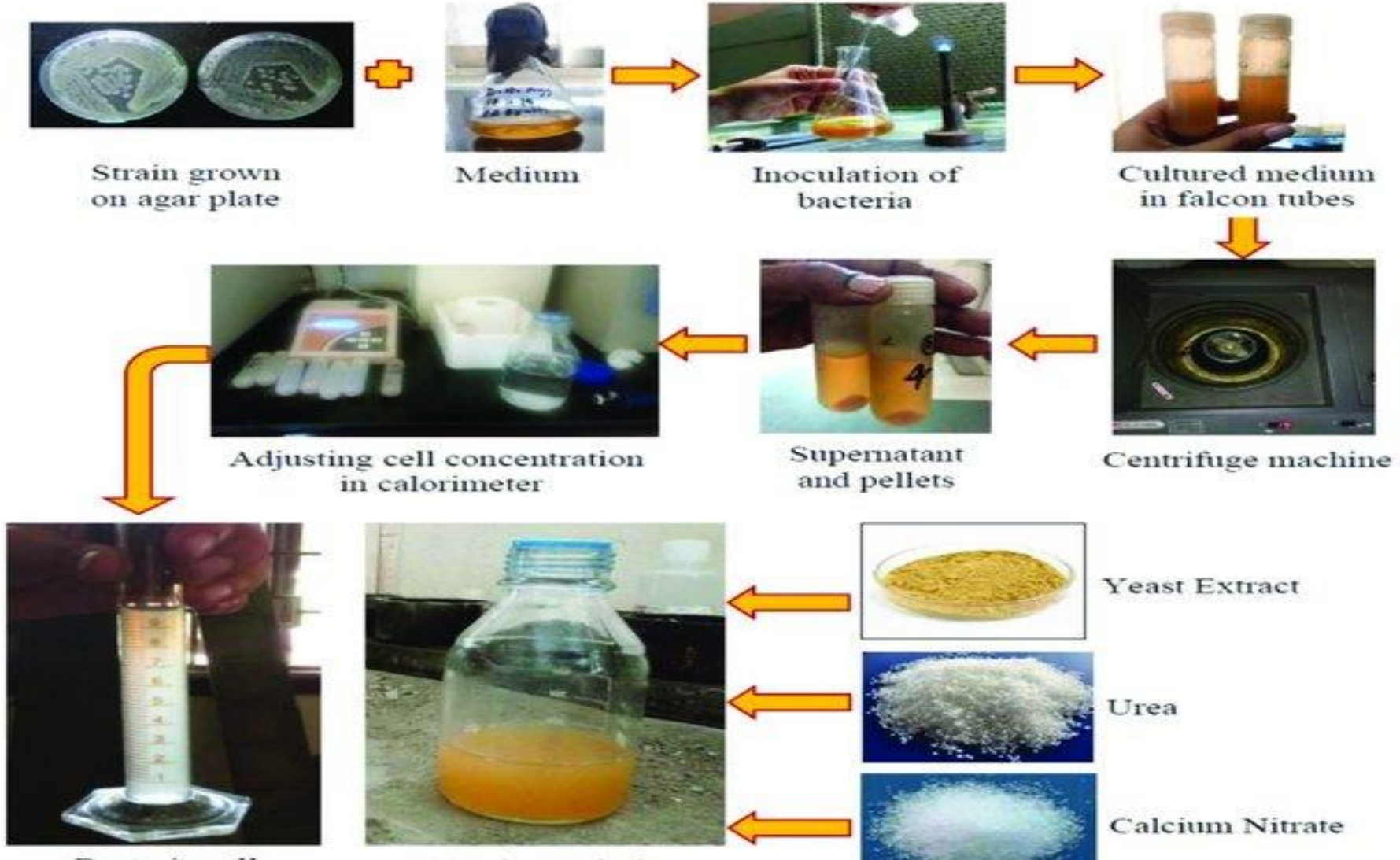


(b)



Indirect Methods – measure effects of bacterial growth

Dry Weight



Metabolic Activity

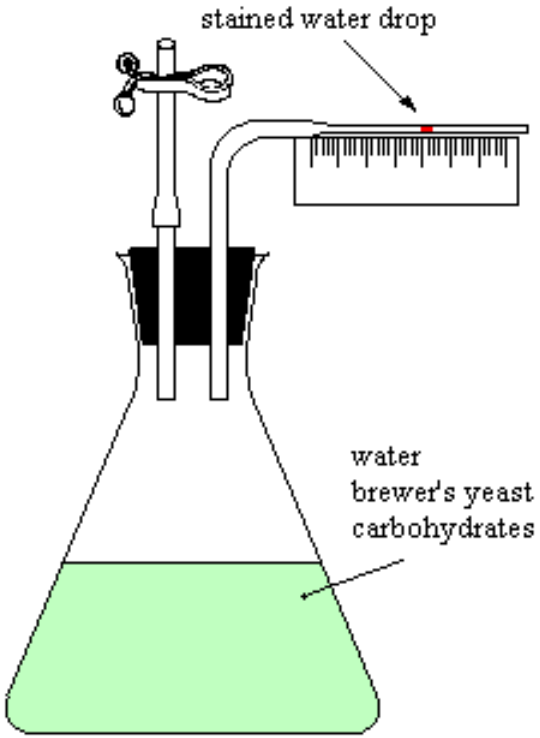


Figure 18 - Apparatus for measuring the production of carbon dioxide during fermentation.